

A MILP Framework for Multi-Product Vehicle Routing Problem with Changeover Costs



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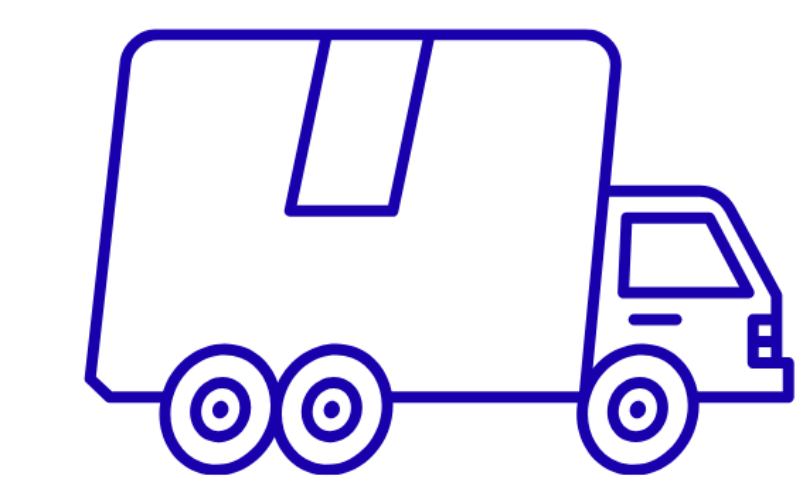


Abstract

Optimizing distribution routes remains a major challenge for logistics companies, particularly in the transport of petroleum, chemical, and food products. We introduce the Multi-Product Vehicle Routing Problem with Changeover Cost (MPVRP-CC), a variant of the Vehicle Routing Problem where a heterogeneous fleet of vehicles trucks must deliver multiple product types from several depots to a network of service stations, balancing the trade-off between making detours to retain the same product and incurring costs for on-site maintenance during product changeovers.

We present (i) a Mixed-Integer Linear Programming (MILP) formulation of the problem, (ii) a public platform providing access to benchmark instances and visualization tools, and (iii) experimental results based on 150 instances with diverse characteristics.

Problem formulation



- A fleet of vehicles. Each
- starts at known garage;
 - has a different load capacity;
 - can hold only one product at a time.



A set of multiple products to distribute.

A trip = Garage + Depot + [Station, ...] + Depot + [Station, ...] + ... + Garage

Valid solution if :

- Total demand covered
- Split-deliveries /single-truck visits
- Garage to garage routing
- No vehicle overload

Stations/customers made requests for differents products.
Depots (loading points) hold an amount of products in stocks.

Objective function

$$\text{Min}(\text{Map} + \text{Cost})$$

Figure 1. Illustration of the MPVRP-CC logistic network.

MILP model

We formalize the MPVRP-CC as a Mixed-Integer Linear Programming (MILP) model and solve it using **Gurobi** optimization solver.

The maximum number of mini-routes R per vehicle is bounded by:

$$R = \max \left(\left\lceil \frac{\sum_{s \in S} \sum_{p \in P} H_{sp}}{\sum_{k \in K} Q_k} \right\rceil, |P| \right)$$

where H_{sp} is the demand of station s for product p , $|P|$ the number of product and Q_k is the capacity of vehicle k .

The model allows split deliveries: a station's demand for a product may be served by several vehicles, provided that one vehicle does not visit the same station twice for the same product.

Objective Function

The objective aims to minimize the total operational cost:

$$\begin{aligned} \min & \sum \text{travel distances (mini-routes + garage access/return)} \\ & + \sum \text{empty transfers between depots} \\ & + \sum_{k \in K} \sum_t \sum_{p \in P} \sum_{p' \in P} C_{pp'} z_{ktpp'} \quad (\text{changeover cost}) \end{aligned}$$

Key constraints

- **demand satisfaction:** All positive station-product demands must be exactly satisfied, supporting split deliveries: $\sum_k \sum_t q_{ktsp} = H_{sp}$.
- **capacity & flow:** The total load transported in a mini-route cannot exceed vehicle capacity Q_k . Subtour elimination is strictly enforced via Miller-Tucker-Zemlin variables.
- **uniqueness:** A vehicle cannot visit the same station more than once for the same product during its route.
- **changeover logic:** A cost penalty $C_{pp'}$ is applied if consecutive mini-routes carry different products p and p' .

Contact us



For any inquiries or potential collaboration opportunities, please feel free to reach out to our team:
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Contribution

To foster further research and reproducibility, we have deployed a public platform for the MPVRP-CC project available at:

<https://ifri-ai-classes.github.io/MPVRP-CC/index.html>

The platform provides:

- Benchmark instances modeling realistic logistics networks.
- Interactive visualization tools for exploring routes and depot states..
- Instance generation tools to create custom scenarios.
- Feasibility checker to validate your solutions.
- Leaderboard and submission portal for competitive benchmarking.

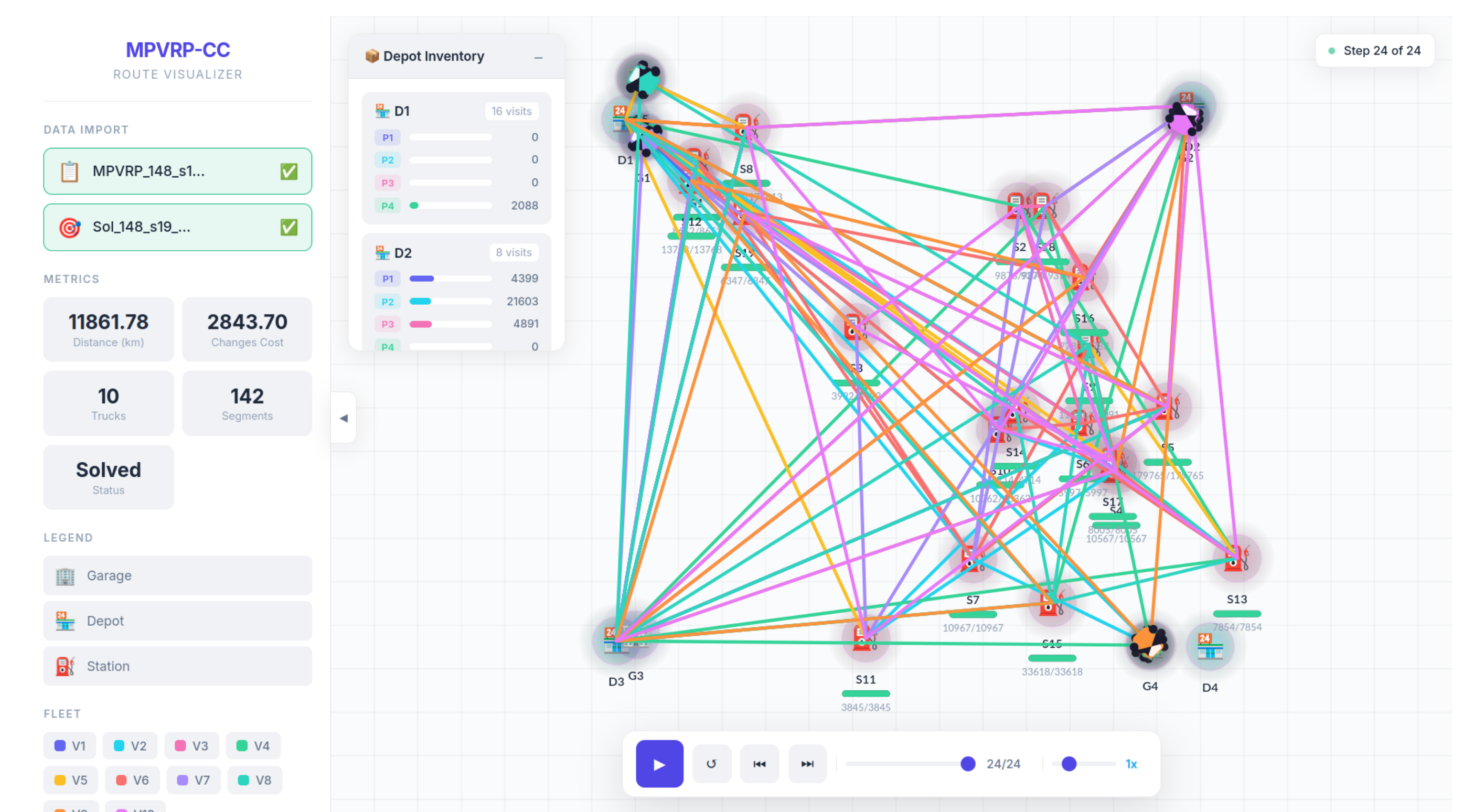


Figure 2. Screenshot of the MPVRP-CC public platform.

Results

The proposed model was evaluated to demonstrate its robustness and structural validity.

- **Dataset:** We worked on **150 instances**.
- **Characteristics:** The instances feature varied and diverse characteristics, evaluating the model's behavior across different fleet configurations, demand asymmetries, and depot network topologies.



Figure 3. Performance metrics across different instance sizes.

Perspectives

The project is currently in an active academic research phase, with the ambition of transitioning into a concrete industrial application:

- identifying a specific business context and establishing a partnership with a logistics enterprise to apply the model in real-world operational scenarios;
- design and implement more models using constraint programming and metaheuristics to efficiently solve large-scale instance;
- launching an open optimization competition based on our platform to invite the global Operations Research community to propose novel solution algorithms.

References

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